

Course “Advanced Topics in Banking”
Summer Term 2026

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Case Study: Credit Risk Measurement

Task:

Your task is to measure the credit risk for a given credit portfolio. For this purpose, you have been given an Excel-Sheet that contains two pieces of information. The first data is a rating history of 10,000 representative borrowers and the second data consists of a credit portfolio of 100 customers for which the credit risk has to be measured. The credit risk measurement is to be carried out in different ways in order to discuss the differences in the results and the advantages and disadvantages. You are to proceed as follows:

1. Based on the rating history of 10,000 representative borrowers, determine the one-year migration matrix based on the Lando and Skodeberg procedure. In doing so, determine the migration matrix using the generator matrix based on at least the first five terms.
2. Determine the credit risk for the portfolio of 100 borrowers based on the Basel Standardized Approach. Proceed by using the existing ratings as if they were external ratings.
3. Determine the credit risk for the portfolio of 100 borrowers based on the Basel IRB Approach based on the regulatory corporate formula. Use the PDs from step 1 (where the minimum PD of a rating class should equal 5 bp), set the LGDs to 55%, and set the maturity to 2.5 years.
4. Determine the credit risk for the portfolio of 100 borrowers based on a simulation analysis. This simulation has to be implemented based on the 2-factor Merton model and has to be based on at least 5,000 scenarios. Let the change in value of each firm depend on a common market factor and an idiosyncratic factor. Use the PDs from step 1 (where the minimum PD of a rating class should equal 5 bp) and set the LGDs to 55%. Set the weight used to model the two factors to 0.37 ($w = 0.37$). Determine the unexpected loss based on the 99.9% VaR (0.1% quantile of the loss distribution - expected loss).
5. Determine the credit risk for the portfolio of 100 borrowers based on a simulation analysis. This simulation has to be implemented based on the 2-factor Merton model and has to be based on at least 5,000 scenarios. Assume here that the change in value of each company depends on an industry factor and an idiosyncratic factor. This means that companies in one industry exhibit dependence. Companies in different industries do not exhibit dependence. (Thus, there are three market factors. However, only one of these factors is relevant for each company - depending on the industry). Use the PDs from step 1 (where the minimum PD of a rating class should equal 5 bp) and set the LGDs to 55%. Set the weight to model the dependence between the respective industry factor and the idiosyncratic factor to 0.46 ($w = 0.46$). Determine the unexpected loss based on the 99.9% VaR (0.1% quantile of the loss distribution - expected loss).

Deliverables?

You have to submit two files:

1. R-Program / Excel-Sheet with all the calculations, structured in a clear and understandable way.
2. A presentation that explains how the calculations were performed, presents the results, and analyzes the results. The following aspects have to be addressed in particular: The different measurement methods for credit risk produce different results. Why do the results differ? What are the advantages and disadvantages of the methods and what is their impact? Which method(s) should be preferred?

The two files have to be uploaded to Canvas till May 24, 2026, 23:59.